Preactivation and Probabilistic Inference Coexist during Sentence Comprehension

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One of the most robust findings in human sentence comprehension is that contextually unpredictable words have larger processing difficulty as measured by behavior, neuroimaging, and electrophysiology. However, what these predictability effects reveal about sentence comprehension is not yet settled. Under the ***facilitation view***, predictability facilitates processing through anticipatory preactivation of likely next words, yielding linear predictability effects (Brothers and Kuperberg, 2021). Under the ***cost view***, predictability effects reflect probabilistic inference over many possible interpretations of the unfolding sentence, with cost proportional to the word's Shannon information or *surprisal*, a logarithmic function of predictability (Smith & Levy, 2013). Conceptually, preactivation and probabilistic inference may coexist: comprehenders could both perform probabilistic inference over many alternative interpretations and preactivate upcoming words when they are highly predictable. This would lead to a superposition of linear and logarithmic components. Several prior high-powered studies have produced discrepant findings: naturalistic reading studies using statistical (language model, *LM*) predictability estimates supported a logarithmic but not a linear effect (Shain et al., 2024), a controlled reading study using cloze predictability estimates supported a linear but not a logarithmic effect (Brothers & Kuperberg, 2021), and controlled studies of the N400 event-related potential (ERP) supported the coexistence of both effects under both cloze and statistical predictability (Szewczyk & Federmeier, 2022; Michaelov & Bergen, 2024). Here we tested a possible explanation of this discrepancy: both linear and logarithmic effects may coexist in reading, but the linear effect could have been masked in naturalistic reading because high contextual constraint is rare, whereas the logarithmic effect could have been masked in Brothers & Kuperberg's analysis because cloze estimates capture it poorly (due to low precision of small cloze probabilities).

Our approach was simple: we ran an online replication of Brothers and Kuperberg's English-language self-paced reading experiment (n=981 participants) and cloze-norming task (n=915 participants) at sufficient scale to identify subtle nonlinearities in the functional form of predictability effects. We used the same 648-sentence bank with critical words normed for high, moderate, and low cloze probability. This design oversamples high constraint items relative to typical text, which is desirable given our goal of detecting changes in the response in precisely such contexts. Participants read one of three counterbalanced lists of 216 sentences with comprehension checks in between. Predictability effects in the three-word critical region were visualized using generalized additive models and tested using predictive fit of linear mixed-effects models under 5-fold cross-validation. In addition to cloze, we considered predictability estimates from seven different LMs (GPT-2, GPT-2-XL, GPT-Neo, GPT-NeoX, GPT-J, OLMO-2, LLaMA-2) on both probability and surprisal metrics.

We found that, across predictability estimators, reading times were best explained by a weighted combination of linear and logarithmic predictors, with Cloze predictability showing a more linear effect and LM predictability showing a more logarithmic one (Fig 1). But, importantly, effects were not purely linear or logarithmic: all models were visually nonlinear on both scales (Fig 1), and all but one model generalized significantly better when both scales were included (Tab 1). These findings add behavioral support for the hypothesis, until now based solely on the N400, that preactivation and probabilistic inference coexist and differentially modulate processing demand: highly predictable words result in facilitation beyond that expected under the cost view, and highly unpredictable words result in inhibition beyond that expected under the facilitation view. This outcome helps reconcile puzzling discrepancies in the field both between different behavioral studies and between behavioral and ERP work, by advancing a unified picture in which both processes are systematically at work but are differentially enhanced by study design details (naturalistic vs. controlled, LM vs. cloze).

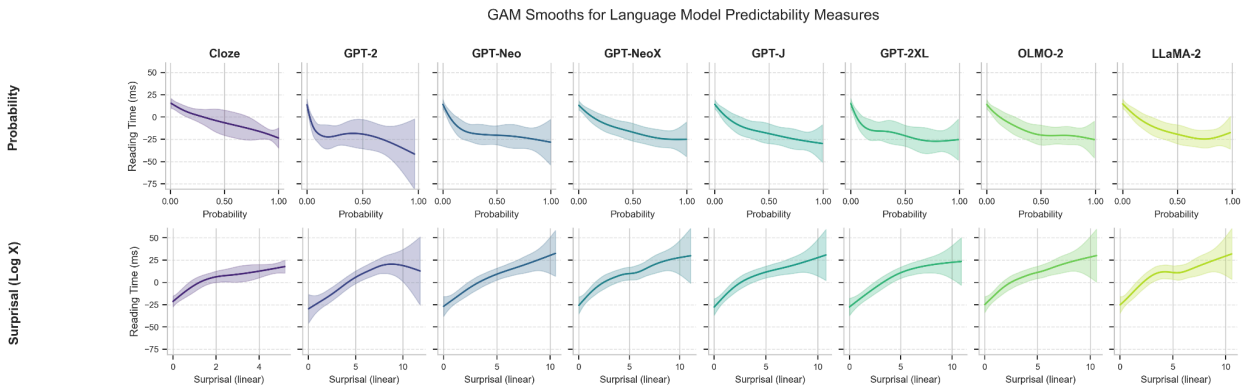

Figure 1: GAM-estimated functional form of effects in SPR experiment across cloze and 7 different language model types (GPT-2, GPT-Neo, GPT-NeoX, GPT-J, GPT-2XL, OLMO-2, LLAMA-2)

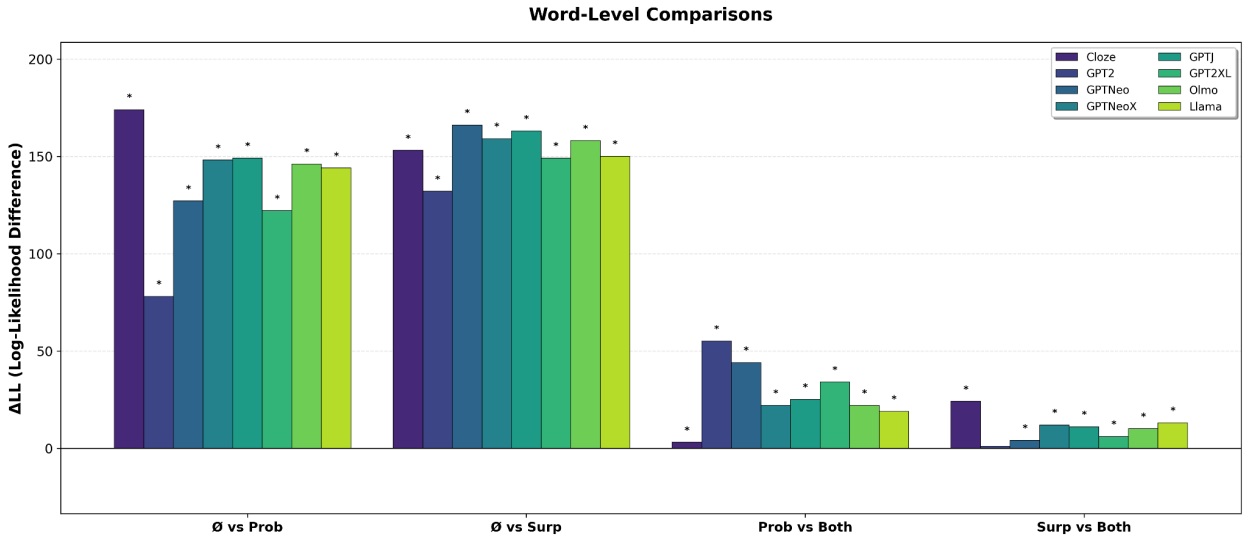

Table 1: Results of key comparisons using paired permutation tests of the difference in cross-validated conditional log likelihood (ΔLL) between linear mixed effects (LME) models.

References: Brothers & Kuperberg 2021; Kretzschmar et al 2015; Ryskin & Nieuwland 2023; Shain et al 2020; Shain et al 2024, *PNAS*; Smith & Levy 2011; Smith & Levy 2013; Staub et al 2015; Szewczyk & Federmeier 2022